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Project Title

UMUGANDA MANAGEMENT SYSTEM

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List of Abbreviations

Abbreviation	Meaning
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
ERD	Entity Relationship Diagram
GPS	Global Positioning System
HTML	HyperText Markup Language
ICT	Information and Communication Technology
JS	JavaScript
JSON	JavaScript Object Notation
MoMo	Mobile Money
PDF	Portable Document Format
PWA	Progressive Web Application
QR	Quick Response (code)
RSVP	Répondez S'il Vous Plaît (Please Respond)
RWF	Rwandan Franc
SDG	Sustainable Development Goal
UI	User Interface
UR-CST	University of Rwanda - College of Science and Technology
UX	User Experience

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Umuganda is a longstanding Rwandan tradition of community work that has evolved into one of the most recognisable expressions of civic responsibility in modern Rwanda. The word Umuganda comes from the Kinyarwanda root -ganda, meaning "coming together in common purpose to achieve an outcome." Historically, families and neighbours would join efforts to build houses, clear fields, or help each other in times of need.

After the 1994 Genocide against the Tutsi, the Government of Rwanda formally revived Umuganda as a national civic activity to rebuild social cohesion, promote unity, and mobilise citizens around shared development goals. Since 2009, Umuganda has been officially organised on the last Saturday of every month between 8:00 AM and 11:00 AM. Every Rwandan citizen aged 18 to 65 is expected to participate in cleaning streets, building schools and health centres, repairing infrastructure, planting trees, or supporting vulnerable members of the community.

Umuganda is coordinated at the umudugudu (village) level, which is the smallest administrative unit in Rwanda. The village leader (umukuru w'umudugudu) is responsible for mobilising villagers, planning the activity for the month, verifying attendance, and reporting to higher administrative levels. Local government inspectors periodically oversee these activities to ensure fairness and impact.

Although the practice remains highly successful, day-to-day management is still largely paper-based. Attendance is often recorded in notebooks, absentee fines are collected informally, and communication between leaders and villagers depends on physical announcements or word of mouth. This creates administrative inefficiencies, opportunities for disputes, and difficulty in producing accurate reports for higher authorities. The Umuganda Management System has been developed to address these challenges by digitising the entire workflow of monthly community work.

1.2. Problem Statement

Despite Umuganda being one of Rwanda's most impactful civic activities, its administration at the grassroots level still faces several problems:

- Attendance registration is done manually in notebooks. Records are easily lost, misread, or falsified.
- There is no reliable way to verify whether a person actually appeared at the activity site as opposed to being marked present by association.
- Villagers who cannot attend often have no formal channel to inform leaders in advance or to justify their absence.
- Fines for missing Umuganda are collected informally in cash, with no receipts or audit trail. This exposes both leaders and villagers to disputes.
- Announcements from leaders about upcoming activities are rarely reaching everyone on time.
- Inspectors depend on written reports that are slow to compile and difficult to analyse for trends across many villages.
- Villagers have no formal way to propose ideas for future community projects.

The absence of a digital, transparent and multilingual platform slows down decision-making, discourages participation, and reduces accountability. This project proposes a modern web-based solution designed specifically for the Rwandan context.

1.3. Objectives of the Project

1.3.1 General Objective

The general objective of this project is to design and develop a fully responsive, mobile-first web application that digitises the entire Umuganda workflow, from planning through attendance verification to reporting, while remaining accessible to communities with limited internet connectivity.

1.3.2 Specific Objectives

- To provide a Kinyarwanda and English user interface so that all Rwandans, regardless of language preference, can use the system comfortably.
- To allow villagers to confirm attendance or provide a documented reason for their absence in advance of each Umuganda.
- To equip leaders with a cross-check tool that compares villagers' declared availability with actual attendance and automatically issues absence charges.
- To offer two on-site check-in methods (a leader-generated QR code and GPS-based verification) that prevent false attendance claims.
- To digitise the payment of absence fines through MTN Mobile Money references, with a leader approval step for accountability.
- To give inspectors a real-time analytics dashboard including charts, a monthly forecast, and downloadable PDF and CSV reports.
- To ensure the application works offline as a Progressive Web App, so it remains useful even in areas with poor connectivity.
- To make the system inclusive by allowing villagers to submit suggestions for future Umuganda activities.

1.4. Expected Outcomes

After successful implementation of this project, the following outcomes are expected:

- Every villager registered in the system has a private account through which they receive announcements, confirm attendance and view their fines.
- Leaders spend significantly less time on paperwork thanks to automated attendance summaries and one-tap actions.
- Absence charges are transparently issued, tracked, and paid through Mobile Money.
- Inspectors obtain accurate, up-to-date reports without waiting for handwritten summaries from every village.
- Communities benefit from stronger participation because the process becomes fairer, more visible and easier to follow.

- The system serves as a reusable civic-tech platform that could later be extended to other community activities such as parents' meetings or emergency mobilisation.

1.5. Project Rationale (Importance to the Community)

This project is important for several complementary reasons:

First, it directly serves a national programme that involves every adult Rwandan citizen. Any improvement in efficiency multiplies across the country.

Second, by using Kinyarwanda as a first-class language, the system respects the identity and everyday reality of the citizens who will use it. Systems built only in English often exclude the very people they are supposed to serve.

Third, digitising fines through Mobile Money aligns with Rwanda's Vision 2050 goal of becoming a cashless society. It reduces the risks associated with handling cash at the village level.

Fourth, the analytics and reporting features support evidence-based decision-making. Local administrators can identify villages with declining attendance and take corrective action before problems grow.

Finally, by delivering the tool as a Progressive Web Application that works offline, the project acknowledges the connectivity reality of many rural areas. Villagers do not need expensive hardware or constant internet to benefit from it.

Chapter 2. System Analysis and Design

2.1. System Analysis

System analysis was carried out to understand who will use the system, what functions it must perform, and how it must integrate with the existing Umuganda workflow. Three categories of user were identified, each with distinct needs, permissions and expectations.

2.1.1 Intended Users of the Project

The system is designed for three types of users, each with a dedicated dashboard and a strictly enforced access level:

User Role	Who They Are	What They Can Do
Villager	Adult residents aged 18-65 who are expected to participate in monthly Umuganda.	View upcoming activities, confirm attendance or decline with reason, check in on Umuganda day using QR or GPS, view fines, pay through MoMo, submit suggestions, read announcements.
Leader	The umukuru w'umudugudu (village leader) or a cell coordinator responsible for organising monthly Umuganda.	Schedule activities, publish announcements, approve or reject absence reasons, mark attendance, generate QR check-in codes, upload photo evidence, confirm MoMo payments, manage villager profiles, print rosters.
Inspector	Cell or sector-level official who monitors Umuganda activities across many villages.	View system-wide analytics, live attendance charts, next-month forecasts, community leaderboard, activity map. Export monthly reports as PDF or CSV. Read-only access.

Table 2.1: Intended users and their permissions

2.1.2 Intended System Partners

The Umuganda Management System is designed to be used and integrated with the following partner institutions and services:

- Ministry of Local Government (MINALOC) — the national body that oversees Umuganda policy. The system provides aggregated attendance and compliance data that can inform national decision-making.
- District, Sector and Cell administrative offices — these levels coordinate Umuganda between the national programme and the villages. Inspector accounts belong to these offices.
- Umudugudu (village) leadership — the everyday users of the leader dashboard. They benefit from the system's automation of attendance and communication.
- MTN Rwanda and Airtel Rwanda — mobile network operators whose Mobile Money services are used for paying absence fines. The system stores the reference number so that the leader can independently verify a payment.

- Rwanda Information Society Authority (RISA) — provides guidelines on digital inclusion. The system's Kinyarwanda interface and offline PWA support are aligned with RISA priorities.
- Local schools, health centres and cooperatives — often the beneficiaries of Umuganda work. Photo-evidence uploads help these beneficiaries document improvements to their facilities.

2.2. System Design

The design of the system follows a three-layer approach: presentation (what the user sees), logic (how the system behaves) and storage (where information is kept). The following subsections describe each design artefact.

1. High-Level Architecture

The Umuganda Management System is built as a Progressive Web Application. This means that although it looks and behaves like a mobile app, it is delivered through a web browser and can be installed on any smartphone or computer. This architecture was chosen because it works on the wide range of devices found in Rwandan villages, requires no app-store distribution, and continues to function even when there is no internet connection.

At a high level, the architecture is composed of the following elements:

- A client-side application written in HTML, CSS and JavaScript, running entirely in the user's browser.
- A local storage engine based on the browser's localStorage API, used to persist all data (activities, RSVPs, attendance, payments, suggestions, announcements and villager profiles).
- A service worker that caches the application shell and enables full offline operation.
- A translation layer that switches between Kinyarwanda and English at runtime without reloading the page.
- Third-party JavaScript libraries loaded from a CDN for advanced features: Chart.js for analytics visualisations and Leaflet.js for the activity map.
- A responsive design layer that automatically adapts to phones, tablets and desktops.

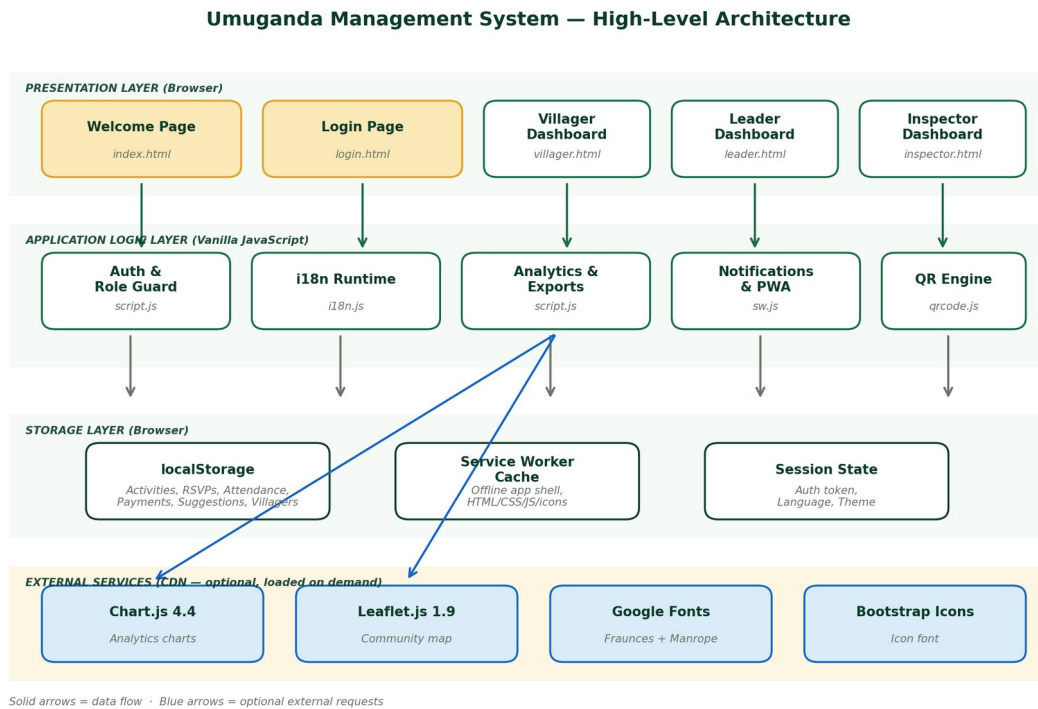


Figure 2.1: High-level architecture of the Umuganda Management System

2. UI Design (Sketch of the UI Module) and Flowcharts

The user interface follows a mobile-first design philosophy. Every screen was first designed for a small phone display and then progressively enhanced for larger screens. This approach ensures that villagers using cheap smartphones have the best possible experience.

The main UI modules of the system are:

- Welcome page: introduces the system and directs users to the login page.
- Login page: allows the user to select their role (Leader, Villager or Inspector) and enter credentials.
- Villager dashboard: shows personal statistics, upcoming activities, attendance history, payments and announcements.
- Leader dashboard: contains the attendance cross-check panel, the QR generation modal, absence-approval requests, payment confirmations, activity scheduling form and villager profiles.
- Inspector dashboard: focuses on analytics with four live charts, a next-Umuganda forecast, a community leaderboard, an interactive map and PDF or CSV report export.

Overall UI Navigation Flow

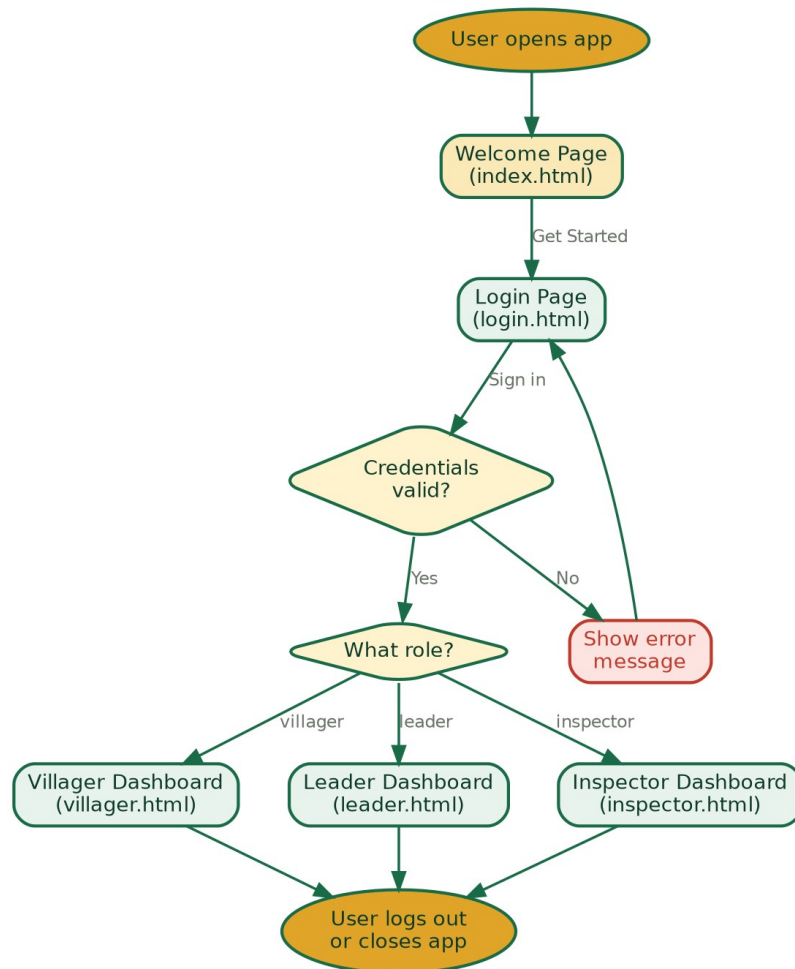


Figure 2.2: Overall navigation flow of the system

Villager Attendance Confirmation & Check-In Flow

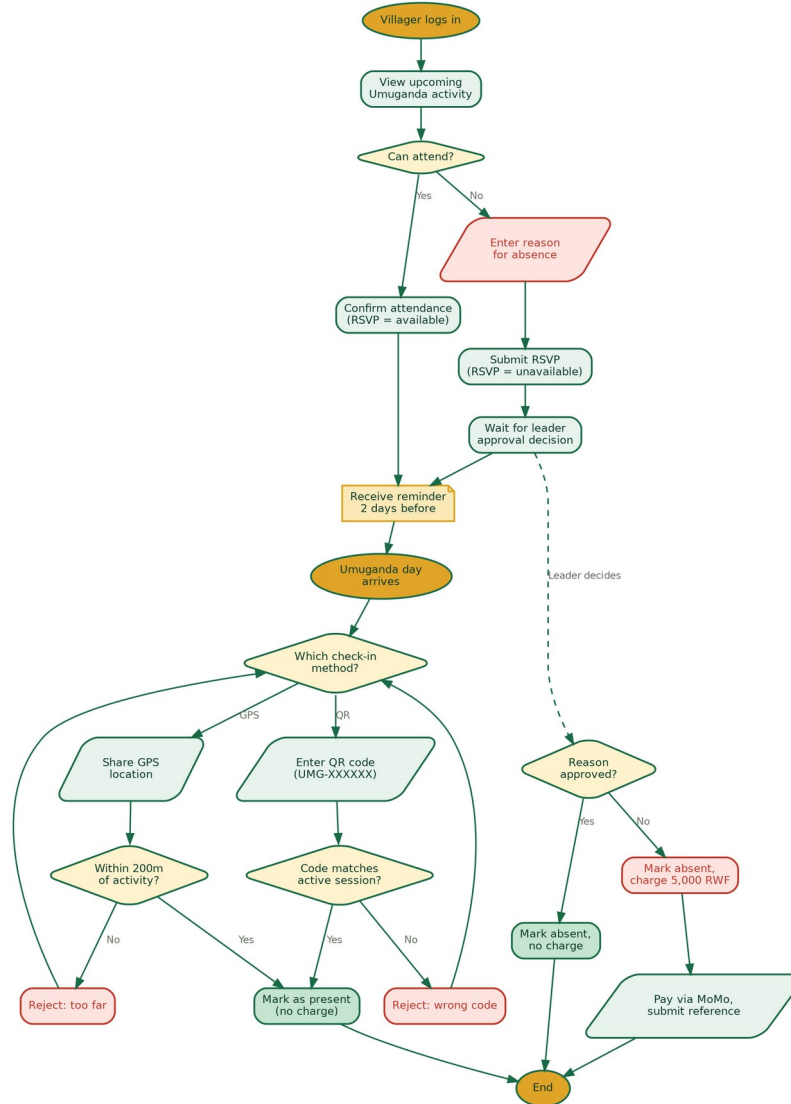


Figure 2.3: Attendance confirmation and check-in flow for the Villager role

Leader Cross-Check & Automatic Charge Issuance Flow

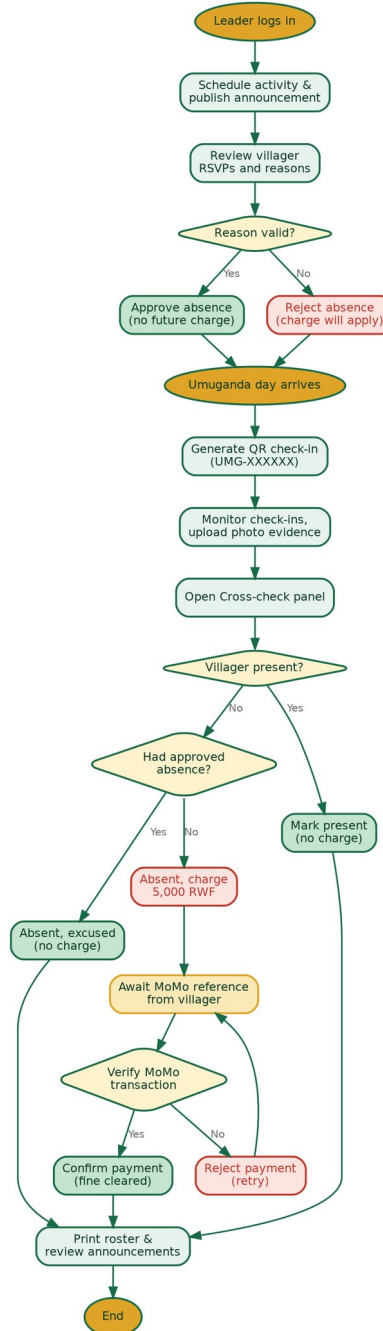


Figure 2.4: Cross-check and automatic charge issuance flow for the Leader role

3. Database Design with ERD and Relationships

Although the system stores data in the browser's localStorage rather than a traditional relational database, the data model itself was designed following the same principles used for relational databases. Every logical "table" in the system corresponds to a JSON collection stored under a dedicated key, and every collection has a well-defined primary key and clearly identified foreign key relationships.

The main entities of the system are:

- User: represents a person authorised to log in. Attributes: username, password, name, role.

- Villager: represents a resident in the community. Attributes: name (primary key), phone, address, profile photo.
- Activity: represents a scheduled Umuganda event. Attributes: id, title, date, location, latitude, longitude, status, photos.
- RSVP: villager's declaration of availability. Attributes: id, activity_id, villager_name, response (available/unavailable), reason, approval_status.
- Attendance: villager's final attendance record. Attributes: id, activity_id, villager_name, status (present/absent), charge, paid, momo_reference, method (leader/qr/gps).
- Payment: villager's fine payment record. Attributes: id, attendance_id, villager_name, amount, momo_reference, status (pending/confirmed/rejected), date.
- Announcement: leader's broadcast message. Attributes: id, title, message, date, author.
- Suggestion: villager's idea for future activities. Attributes: id, message, date, status.

The relationships between these entities are illustrated in the Entity Relationship Diagram below.

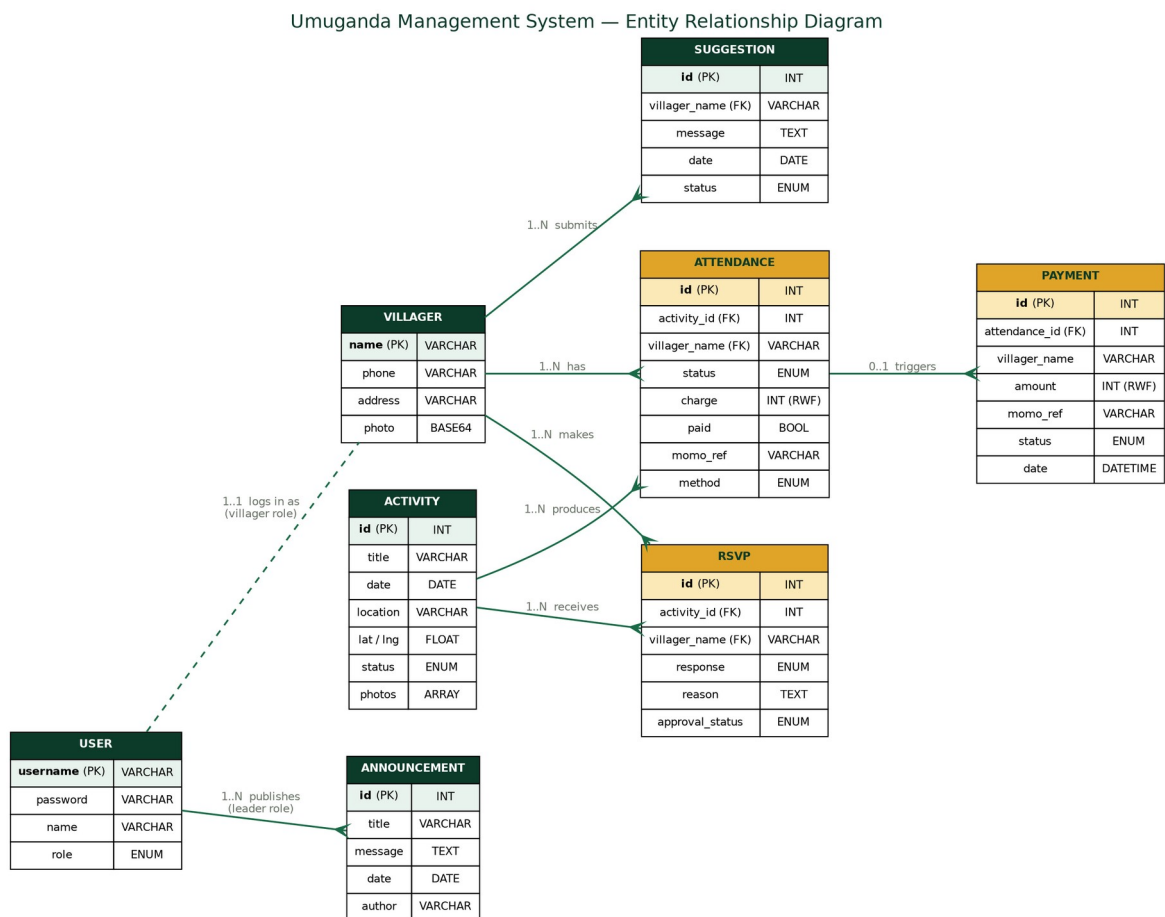


Figure 2.5: Entity Relationship Diagram showing entities and their relationships

Key relationships shown in the ERD:

- One User has one Role (Leader, Villager, or Inspector).
- One Villager can make many RSVPs (one per Activity).
- One Activity can receive many RSVPs (one per Villager).

- One Activity produces many Attendance records (one per Villager who was expected).
- One Attendance record can generate zero or one Payment records (only when charge > 0).
- One Villager can submit many Suggestions.
- One Leader can publish many Announcements.

Chapter 3. Implementation

3.1. Introduction of the Section

This chapter presents the implemented Umuganda Management System through screenshots of each key page and a discussion of what the user sees, what they can do, and how the underlying system responds. All screenshots were taken from the deployed application running in a modern browser.

The system was implemented using the following technology stack:

- HTML5 for structuring content.
- CSS3 with custom design tokens (a self-built design system inspired by Rwanda's green hills and gold sun), used instead of Bootstrap to keep the application lightweight and to demonstrate independent front-end skills.
- Vanilla JavaScript (ES2020+) for all client-side logic. No build tools or frameworks were used; the entire application runs directly in the browser without any transpilation step.
- Chart.js 4.4 for the analytics visualisations on the inspector dashboard.
- Leaflet.js 1.9 for the interactive community map.
- A service worker written in plain JavaScript for offline caching.
- A manifest file plus 192-pixel and 512-pixel icons to make the application installable as a PWA.

Every page enforces its own role guard synchronously in the <head> so that unauthorised users are redirected before any restricted content is rendered.

3.2. Screenshots and Discussion

3.2.1 Welcome Page

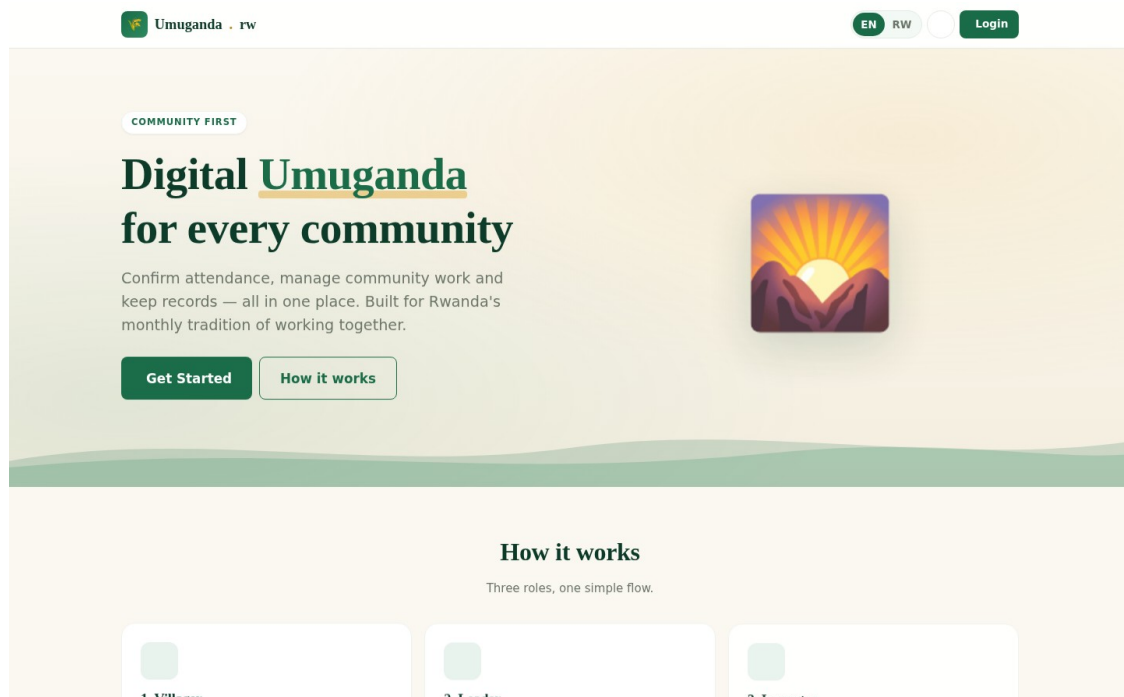


Figure 3.1: Welcome page (index.html)

The welcome page is the public entry point into the system. It opens with a bold hero section featuring the tagline "Digital Umuganda for every community" and two clear calls to action: a green "Get Started" button that leads to the login page and an outlined "How it works" link that scrolls to the explanation section.

Below the hero, a "How it works" section presents the three user roles as visually distinct feature cards. This is followed by an "About Umuganda" section that briefly explains the tradition for any visitor who is not familiar with it, together with headline statistics such as "3 User Roles" and "5,000 RWF Fine".

The top navigation bar includes a language toggle (EN / RW), a dark-mode button, and a login button. The language toggle is one of the most important user-experience decisions of the project because it directly acknowledges that the majority of villagers are more comfortable in Kinyarwanda. When the user taps RW, every text element on the page — including buttons, headings, feature descriptions and footer text — updates immediately without a page reload thanks to the runtime i18n system built specifically for this project.

On small screens, the layout stacks vertically, the hero illustration hides to save space, and the navigation actions collapse to icon-only buttons. This is a direct result of the mobile-first design approach used throughout the project.

3.2.2 Login Page

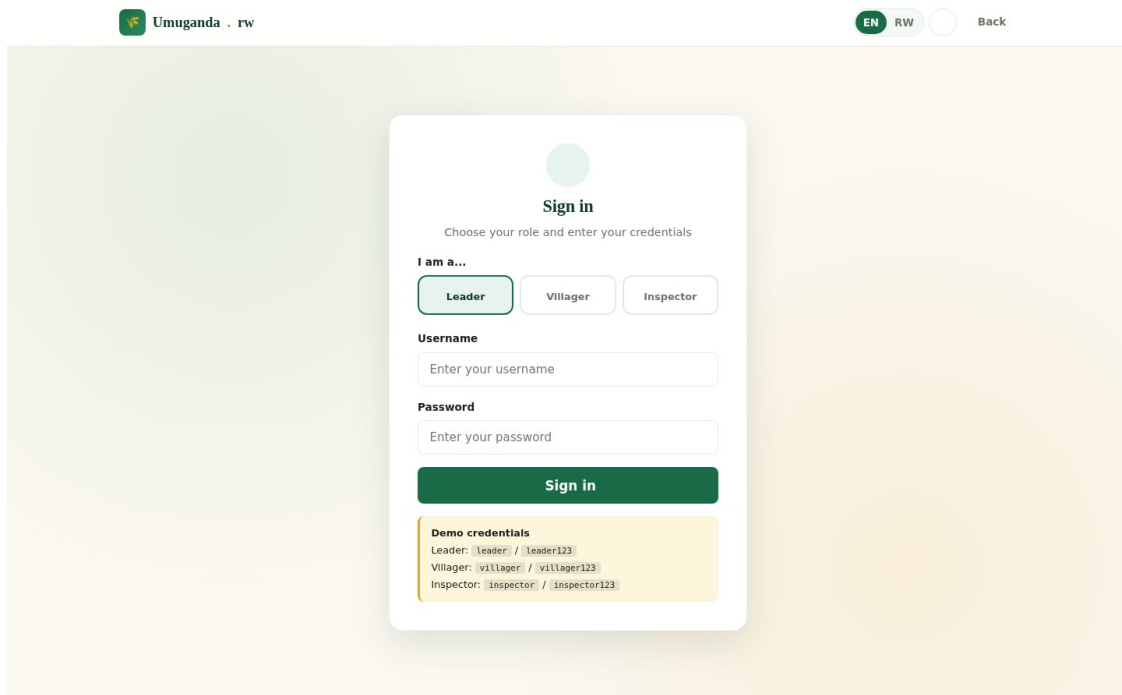


Figure 3.2: Login page (login.html)

The login page presents the user with a clean card centred on the screen. At the top of the card is a lock icon to reinforce the security theme, followed by a role picker with three buttons: Leader, Villager and Inspector. The currently selected role is highlighted with the primary green colour. When the user changes their role, the username field is auto-populated with the role name to make the demo easier to test.

Below the role picker are the username and password fields, followed by a large "Sign in" button. Any invalid credentials trigger a friendly red error banner without navigating away from the page. On successful login, a green toast notification appears at the top of the screen and the user is redirected to the appropriate dashboard after a short delay.

At the bottom of the card is a discreet hint box containing the demo credentials. This is intentional for demonstration purposes and would be removed in a production deployment. The role guard mechanism ensures that if a villager somehow types the URL of the leader dashboard, they will be redirected back to the villager dashboard before any content renders — preventing what is known as a "flash of restricted content".

3.2.3 Villager Dashboard

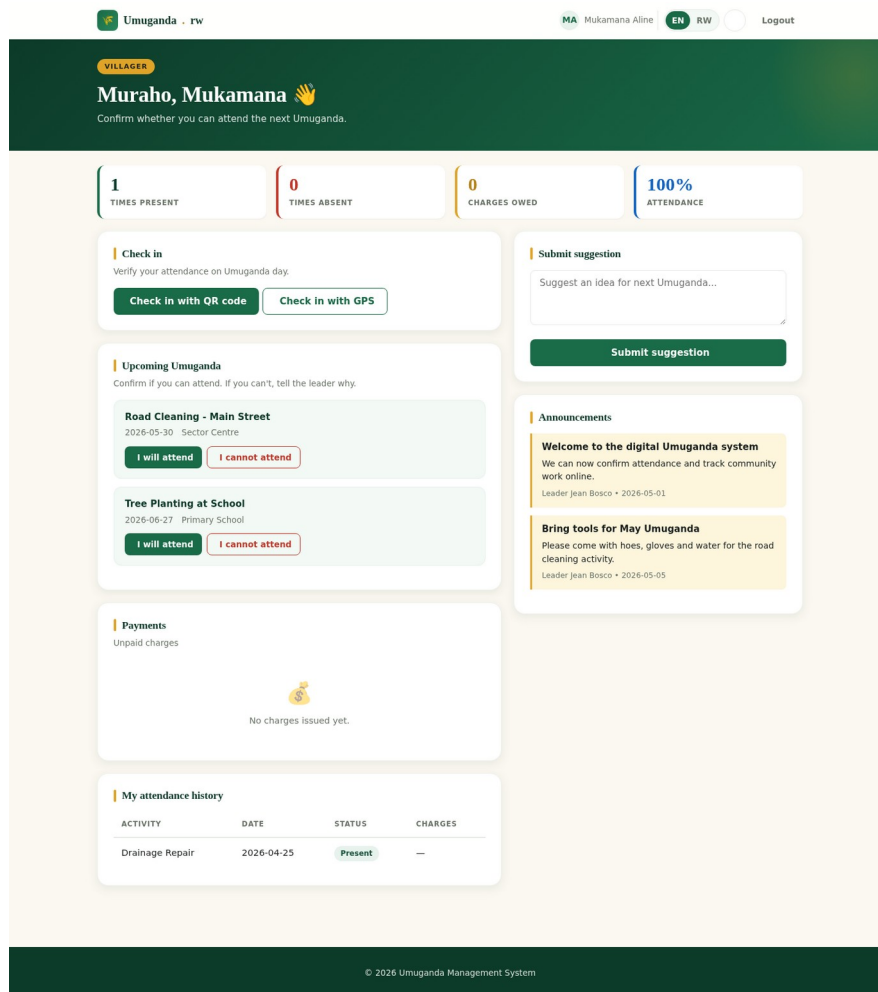


Figure 3.3: Villager dashboard (villager.html)

The villager dashboard is optimised for the everyday needs of a community member. It opens with a personalised greeting ("Muraho, Aline") and four statistics cards at the top: number of times present, times absent, charges owed, and overall attendance rate.

The main content area is split into two columns. On the left, the villager finds a check-in panel offering two verification methods (QR code and GPS), a list of upcoming Umuganda activities with confirm/decline buttons, a payments panel showing any unpaid fines with a "Mark as paid" action, and

a table of attendance history. On the right sidebar are the suggestions form (where villagers can propose ideas for future activities) and the announcements feed from the leader.

An optional notification banner appears if the browser has not yet been granted permission to show reminder notifications. Once the villager enables reminders, the system automatically sends a browser notification two days before every scheduled Umuganda.

On mobile devices, the two-column layout collapses into a single column, and the attendance history table transforms into a set of cards where each cell is labelled with its column name — a technique known as responsive tables. This ensures that the dashboard remains readable and usable on the smallest smartphone screens.

3.2.4 Check-in Modal (QR Code)

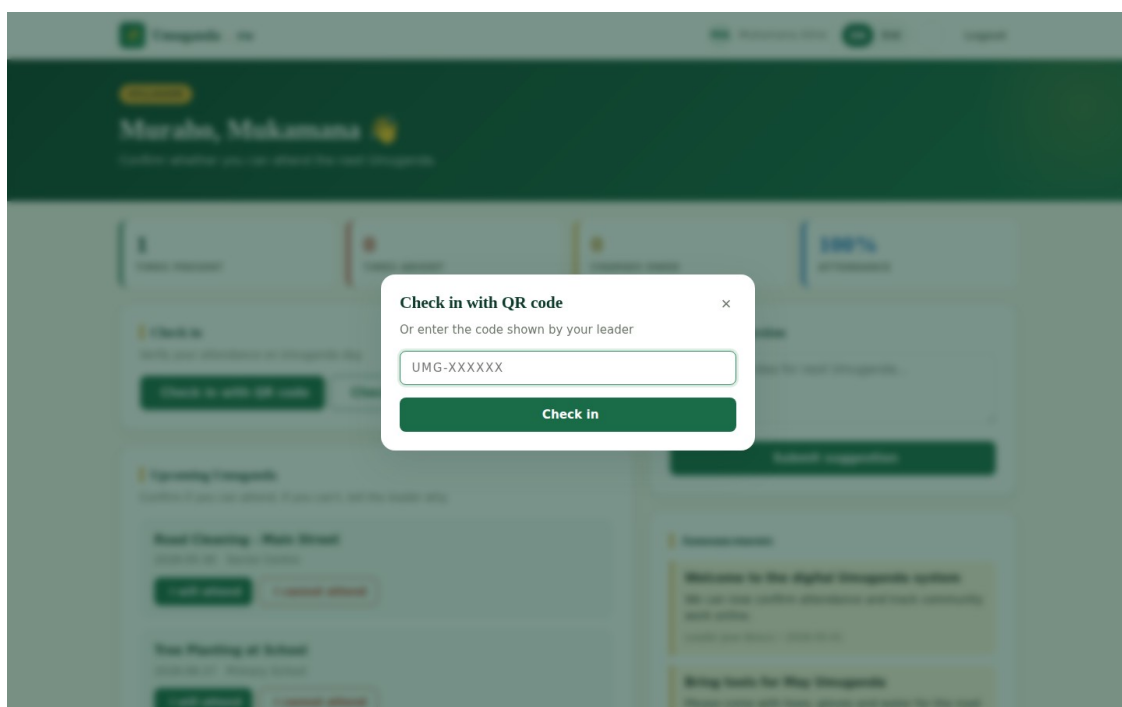


Figure 3.4: QR code check-in modal (villager side)

When the villager taps "Check in with QR code", a modal dialog appears with an input field. The villager either scans the QR code shown by the leader on their own device or types the 6-character code that appears below it (for example, UMG-A7X3P2). If the code matches the active session, the villager is instantly marked present without any further action from the leader. If the code is wrong, a red toast notification informs them that the code is invalid and asks them to check with the leader.

This mechanism prevents two forms of dishonesty. First, a villager cannot mark themselves present remotely because they do not know the code without being at the location. Second, a leader cannot forget to record someone who actually attended because the check-in is automatic. The result is a fair and verifiable attendance process.

3.2.5 GPS Check-in Feedback

The screenshot displays the Umuganda Management System interface for a villager named Muraho, Mukamana. The interface is divided into several sections:

- Header:** Includes the Umuganda logo, a location status "Getting your location...", and user information "MA Mukamana Aline" with language options "EN" and "RW", and a "Logout" button.
- User Profile:** Shows the name "Muraho, Mukamana" and a prompt to "Confirm whether you can attend the next Umuganda."
- Attendance Summary:** Displays four statistics:
 - 1 TIMES PRESENT
 - 0 TIMES ABSENT
 - 0 CHARGES OWED
 - 100% ATTENDANCE
- Check in Section:** Includes a "Check in" prompt to "Verify your attendance on Umuganda day." with two buttons: "Check in with QR code" and "Check in with GPS".
- Submit suggestion Section:** Includes a "Submit suggestion" prompt to "Suggest an idea for next Umuganda..." with a text input field and a "Submit suggestion" button.
- Upcoming Umuganda Section:** Lists upcoming activities:
 - Road Cleaning - Main Street:** 2026-05-30, Sector Centre. Buttons: "I will attend" and "I cannot attend".
 - Tree Planting at School:** 2026-06-27, Primary School. Buttons: "I will attend" and "I cannot attend".
- Announcements Section:** Includes two announcements:
 - Welcome to the digital Umuganda system:** "We can now confirm attendance and track community work online." by Leader Jean Bosco on 2026-05-01.
 - Bring tools for May Umuganda:** "Please come with hoes, gloves and water for the road".

Figure 3.5: GPS-based check-in feedback

As an alternative to the QR code, the villager can tap "Check in with GPS". The browser prompts them to share their location, and the system uses the Haversine formula to calculate the distance between the villager's current position and the recorded location of the activity. If the villager is within 200 metres, the system marks them present and displays a success toast that includes the measured distance (for example, "You are checked in! (47m)"). If the villager is too far away, an error toast informs them of the distance and rejects the check-in.

This location-based verification is especially useful for activities that spread over a wide area, such as road cleaning, where using a single QR code might be inconvenient.

3.2.6 Leader Dashboard

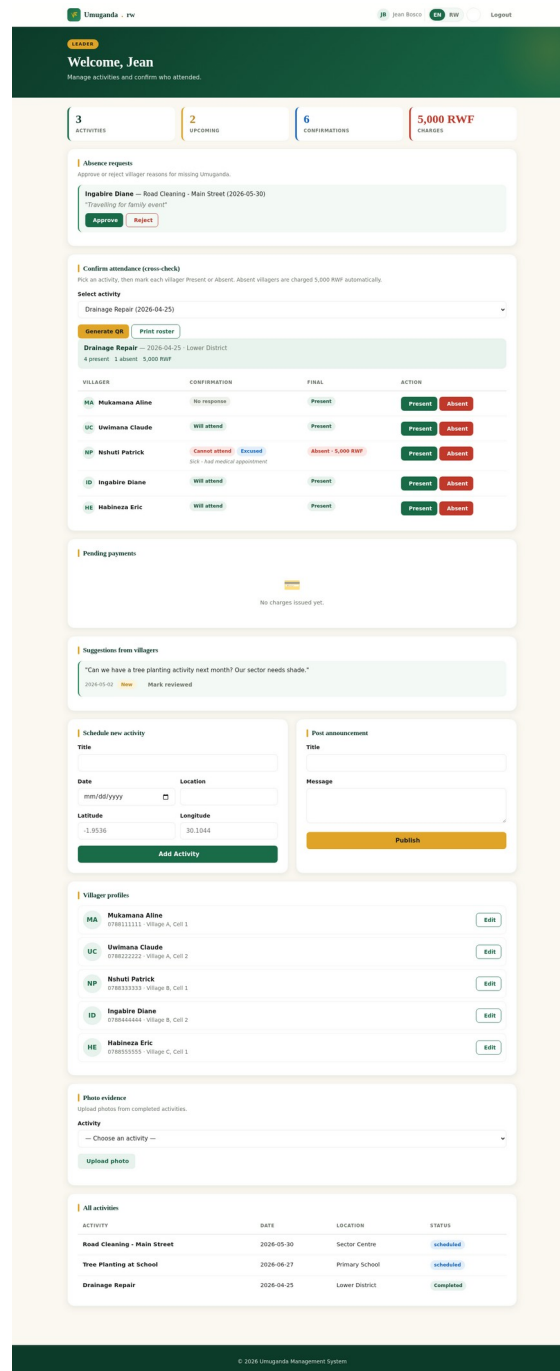


Figure 3.6: Leader dashboard (leader.html) — full-length view

The leader dashboard is the operational heart of the system. It opens with four statistics cards summarising the whole community: number of activities, upcoming activities, confirmations received, and total charges issued. Below the statistics is the absence-requests panel, which highlights villagers who have declined attendance and are waiting for the leader's decision (approve or reject).

The centrepiece of the dashboard is the attendance cross-check panel. The leader selects an activity from a dropdown, then sees a table listing every villager alongside their declared availability, their final attendance status, and any warning flags. For example, a villager who confirmed they would attend but

did not appear is flagged with a red warning; a villager who said they could not attend but came anyway is flagged with a green info message.

Two prominent buttons at the top of this panel let the leader (a) generate a QR check-in session and (b) print the roster as a paper backup. The QR button opens a modal displaying a visual QR-style code and its 6-character text form, both easy for villagers to consume.

3.2.7 QR Code Generation Modal

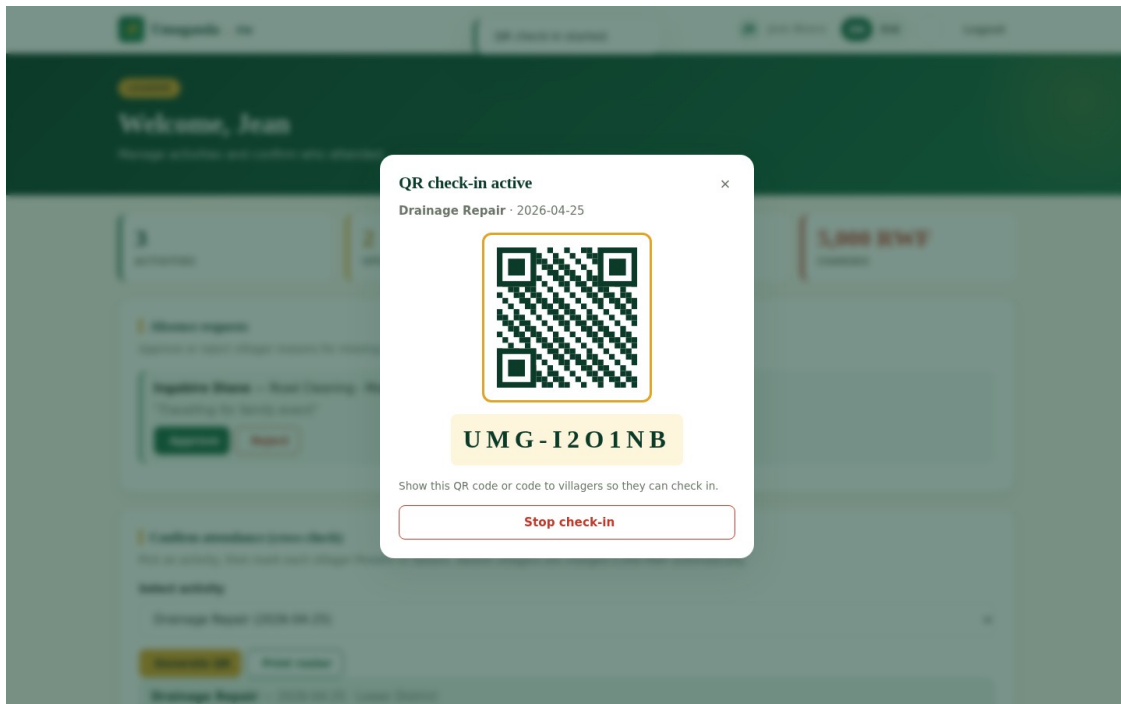


Figure 3.7: Leader generating a QR check-in session

The QR generation modal shows the currently active check-in code along with contextual information about the activity (title, date, location) and a summary sentence: "Show this QR code or code to villagers so they can check in." A red "Stop check-in" button lets the leader end the session at any time. Only one session can be active at a time to prevent confusion.

3.2.8 Payment Confirmation Panel (Leader)

The screenshot shows the Umuganda Management System interface for a leader. The top navigation bar includes the Umuganda logo, the user's name (Jean Bosco), and a Logout button. The main content area is divided into several sections:

- Confirm attendance (cross-check):** A section with a dropdown menu to "Select activity" and a note: "Pick an activity, then mark each villager Present or Absent. Absent villagers are charged 5,000 RWF automatically."
- Pending payments:** A section showing a payment from "Nshuti Patrick" for 5,000 RWF, with MoMo ref: 87654321 - 2026-05-01. It includes buttons for "Confirm payment" and "Reject payment".
- Suggestions from villagers:** A section showing a suggestion from "2026-05-02" with the text "Can we have a tree planting activity next month? Our sector needs shade." and a "Mark reviewed" button.
- Schedule new activity:** A section with a "Title" input field.
- Post announcement:** A section with a "Title" input field.

Figure 3.8: Pending payments awaiting leader confirmation

When a villager submits a MoMo reference number for their unpaid fine, that payment appears in the leader's pending payments panel. The leader can independently verify the payment by checking their own MTN MoMo statement, then click "Confirm payment" or "Reject payment". If confirmed, the villager's fine is marked as paid and the money trail is fully documented. This dual-approval design mimics real-world accounting best practices and protects both parties from disputes.

3.2.9 Villager Profile Editing

The screenshot shows the Umuganda Management System interface with a modal open for editing a villager's profile. The modal is titled "Edit — Mukamana Aline" and contains the following fields and buttons:

- Phone:** A text input field containing "0788111111".
- Address:** A text input field containing "Village A, Cell 1".
- Upload photo:** A button to upload a photo.
- Save:** A green button to save the changes.

Figure 3.9: Leader editing a villager's profile

Every villager has a profile card visible on the leader dashboard, including their name, phone number, address and an optional photograph. When the leader clicks "Edit", a modal appears where the phone number and address can be updated and a new photo can be uploaded. The photo is converted to a Base64 string and stored locally, then displayed as an avatar in the attendance table. This small feature dramatically improves recognition, especially when new families move into the village.

3.2.10 Inspector Dashboard — Analytics Charts

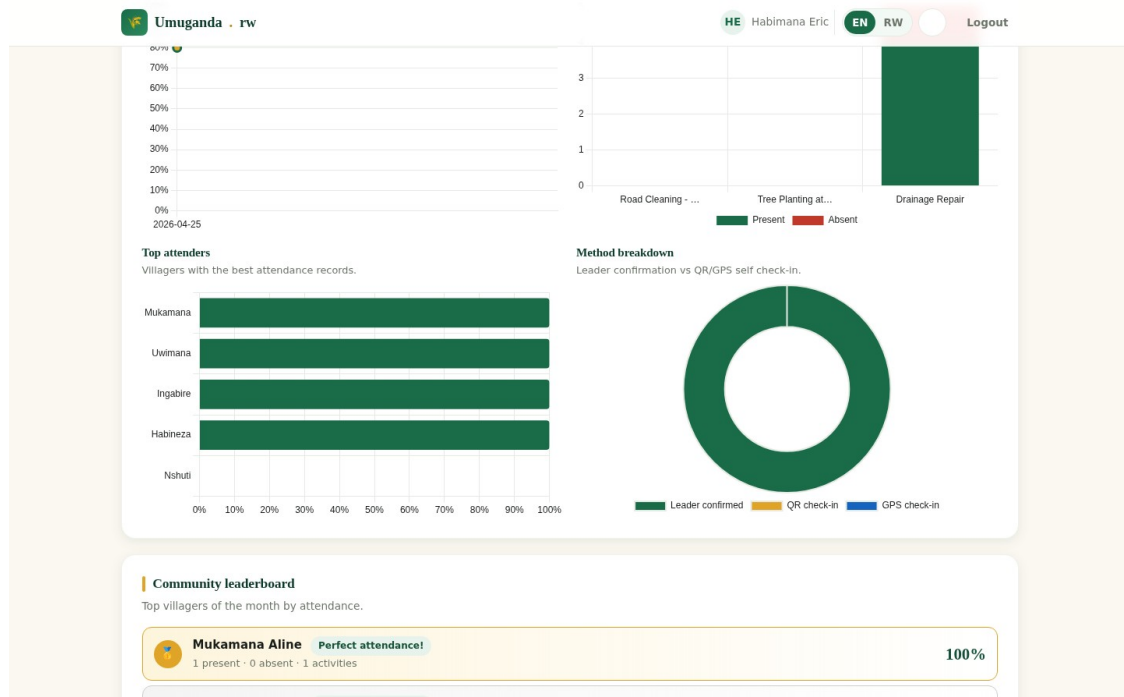


Figure 3.10: Inspector analytics charts (Chart.js)

The inspector dashboard opens with four statistics cards and immediately transitions to the four analytical charts:

- Attendance trend (line chart): shows how the attendance percentage has evolved across every completed Umuganda in chronological order.
- Present vs Absent (stacked bar chart): breaks down each activity into green (present) and red (absent) segments.
- Top attenders (horizontal bar chart): ranks villagers by attendance rate, colour-coded green above 80 percent, gold between 50 and 80 percent, and red below 50 percent.
- Method breakdown (doughnut chart): shows the proportion of check-ins performed by the leader manually, by QR code, or by GPS.

All charts are fully responsive and re-render automatically when the user switches between light and dark mode, thanks to a MutationObserver that watches the data-theme attribute on the root HTML element.

3.2.11 Attendance Forecast

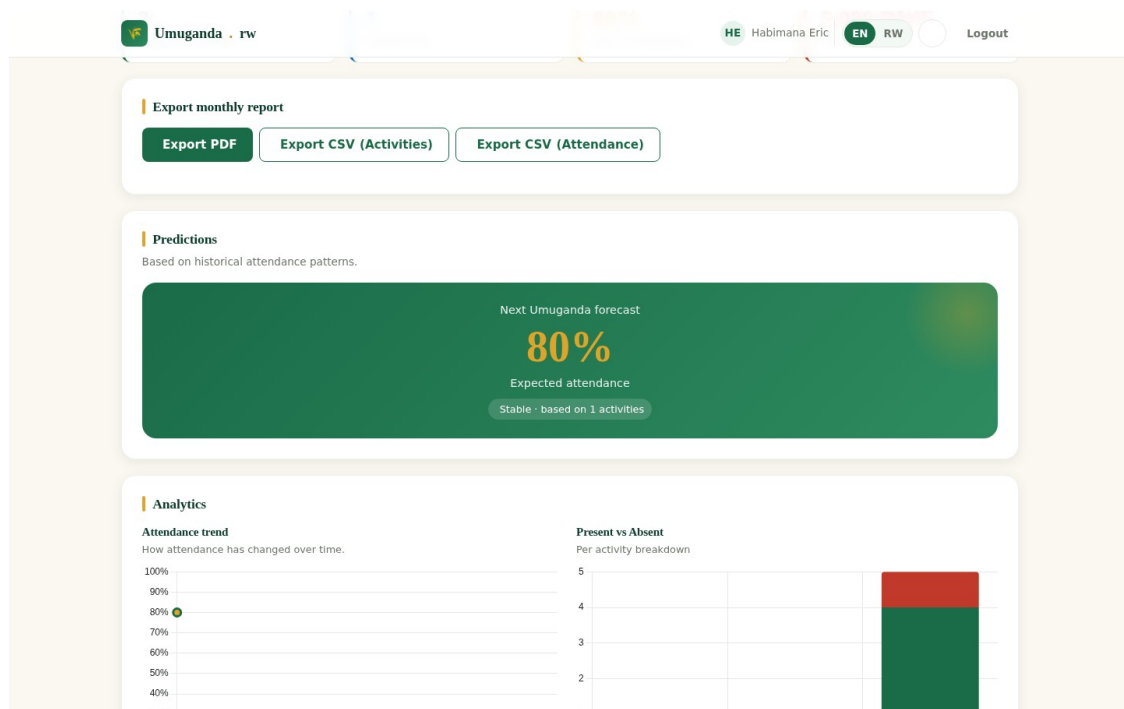


Figure 3.11: Next-Umuganda attendance forecast

Immediately above the charts sits a gradient prediction card. It displays a large percentage (for example, 78%) that represents the expected attendance rate for the next scheduled Umuganda. The prediction is calculated from a rolling average of the last six completed activities plus a simple linear trend adjustment that compares recent activities to older ones. A small arrow indicator (improving, stable, or declining) helps the inspector interpret the forecast at a glance. This is an example of applied analytics that turns raw historical data into actionable insight.

3.2.12 Community Leaderboard

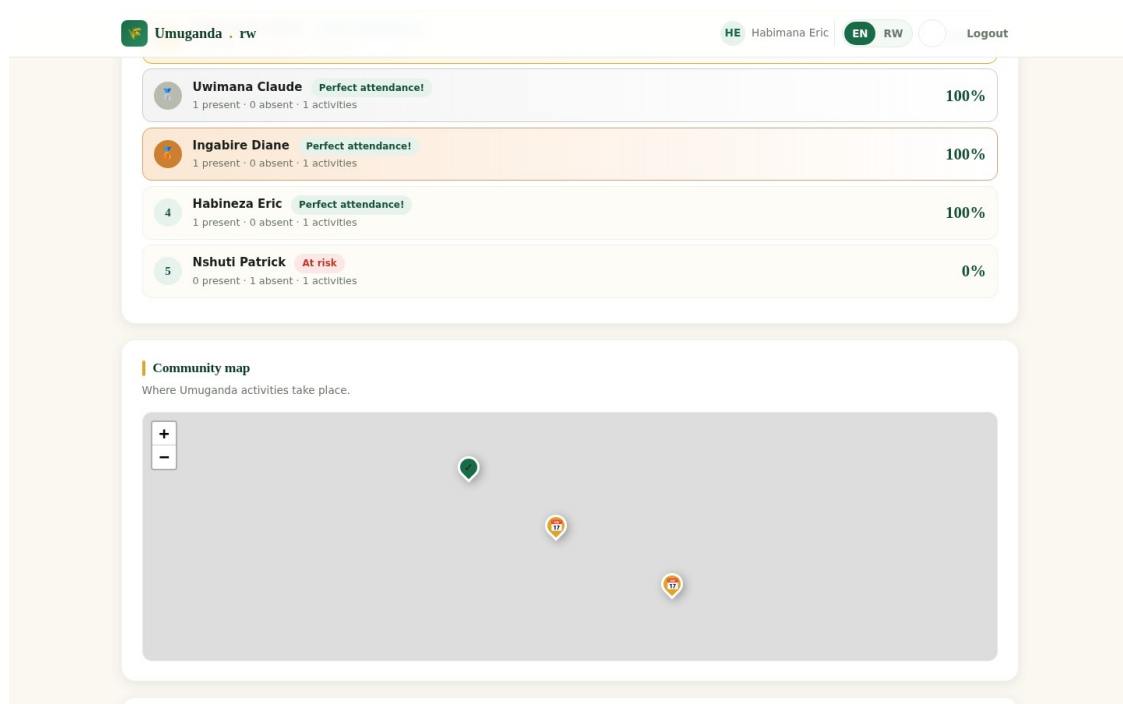


Figure 3.12: Community leaderboard with medals

The leaderboard section ranks every villager by their attendance rate and awards medals to the top three. Villagers with a perfect 100 percent record receive a green "Perfect attendance!" badge, while those below 60 percent are marked "At risk". This gamification element encourages friendly competition and helps inspectors identify villagers who may need additional support or follow-up.

3.2.13 Interactive Community Map

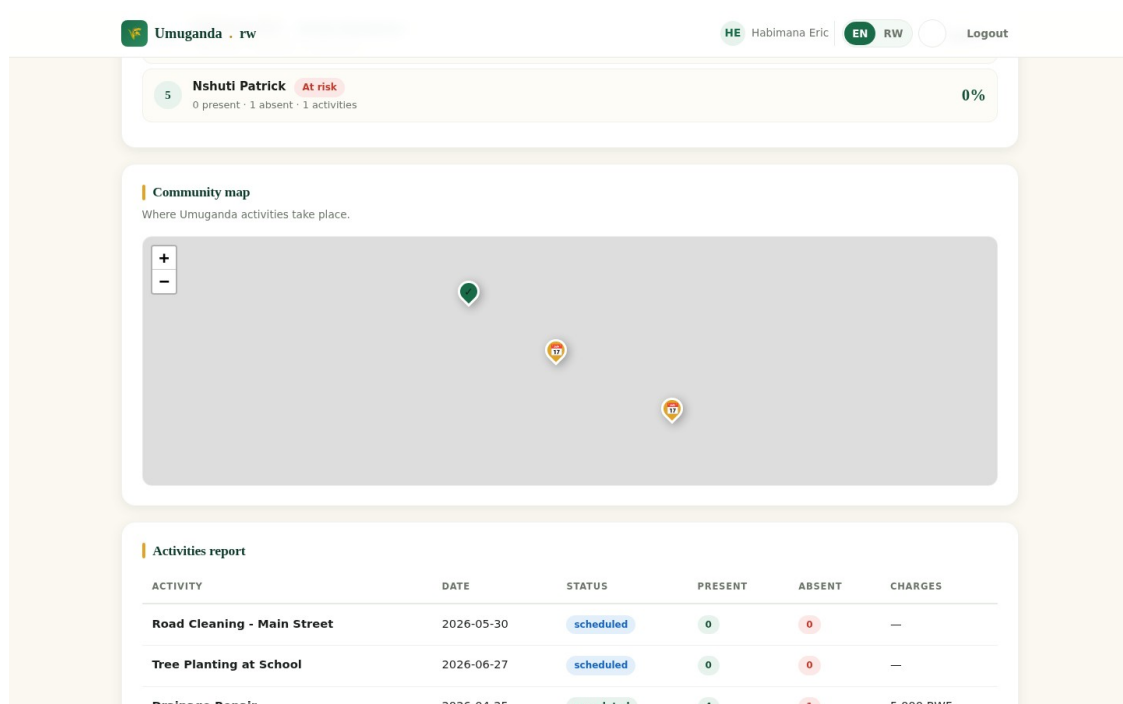


Figure 3.13: Interactive community map (Leaflet.js)

Below the leaderboard is a Leaflet-powered interactive map that plots every activity's location as a custom teardrop marker. Completed activities appear as green markers with a check mark, while scheduled activities appear as gold markers with a calendar icon. Clicking on a marker opens a popup showing the activity's title, date and current attendance count. If the browser is offline, the map area shows a friendly "Map unavailable (no internet)" message instead of a broken tile grid.

3.2.14 Inspector Dashboard — Full View and Report Export

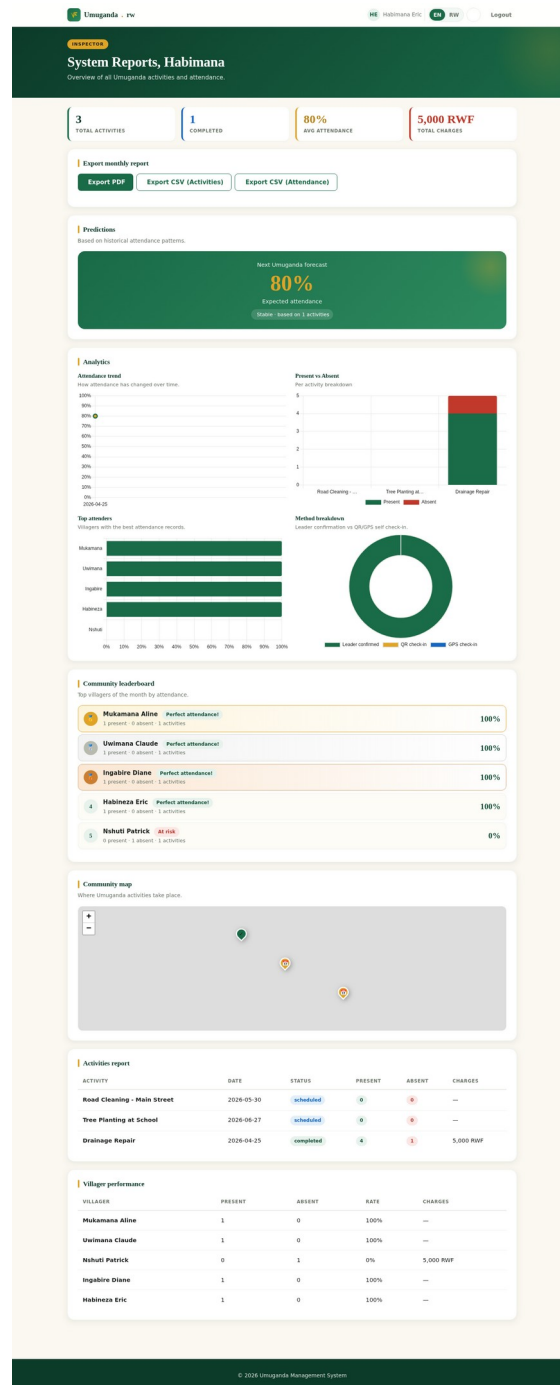


Figure 3.14: Complete inspector dashboard showing all analytics sections

This full-length capture shows the entire inspector dashboard in one view, from the statistics cards at the top through the export toolbar, the prediction card, the four analytics charts, the community

leaderboard, the interactive map, and finally the detailed report tables at the bottom. Everything an inspector needs to monitor Umuganda across a village or cell is available on a single scroll.

The inspector can download three types of report from the export toolbar at the top:

- Monthly PDF report: opens a print-friendly window styled with the Umuganda colour scheme, listing all activities, the community leaderboard, and every charge issued. The user can then save it as a PDF using their browser's print dialog.
- Activities CSV: a spreadsheet-friendly file listing every activity with its date, location, status, present count, absent count, and total charges.
- Attendance CSV: a spreadsheet-friendly file listing every attendance record with its villager, activity, date, status, check-in method, charge, and paid status.

All exports include proper CSV escaping so that names containing commas or quotation marks are handled correctly. The reports are generated entirely in the browser without any server round-trip, making them fast and privacy-preserving.

3.2.15 Dark Mode and Kinyarwanda Language

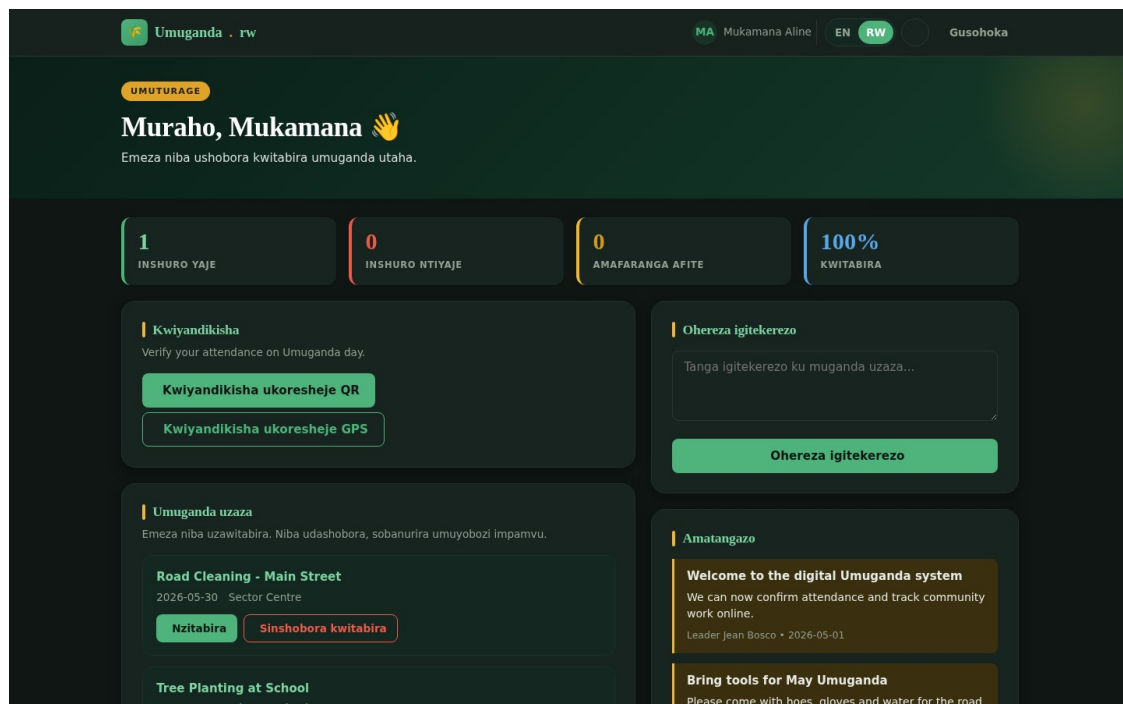


Figure 3.15: The application in dark mode with the Kinyarwanda interface active

Every page in the system supports a dark mode. The theme is applied synchronously before the body renders, so there is no jarring flash when navigating between pages. The user's preferred theme is saved to localStorage and persists across sessions. Combined with the Kinyarwanda language toggle, this gives every user complete control over the appearance and language of the system regardless of their reading preferences or device conditions.

The Kinyarwanda translation is not a machine translation. Every string was carefully worded to match the tone that would be used by a village leader speaking to their community, using culturally

appropriate greetings such as "Muraho" and administrative terms such as "umukuru w'umudugudu" and "kwiwandikisha" (to register).

3.2.16 Mobile Responsive Design

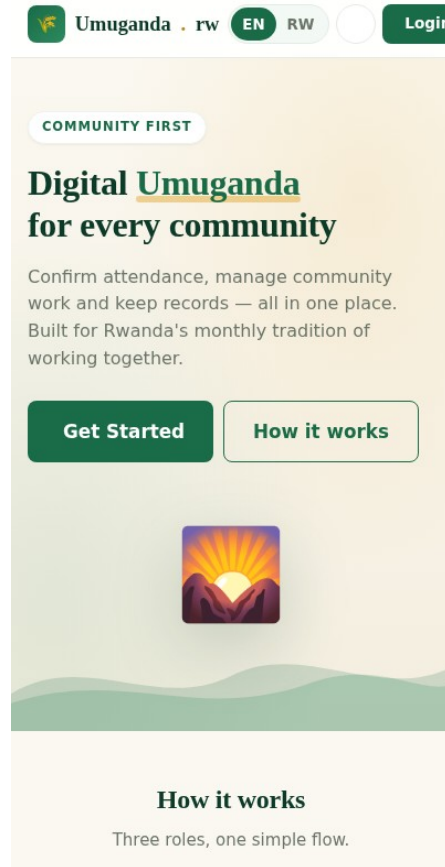


Figure 3.16: Mobile view of the welcome page

Because most Rwandan villagers access the internet through smartphones, the entire application was designed mobile-first. This capture shows the welcome page on a 400-pixel-wide phone viewport. The hero illustration hides to save space, headlines resize to fit narrow screens, and all navigation actions collapse to icon-only buttons. Touch targets are kept above 44 pixels to satisfy accessibility best practices.

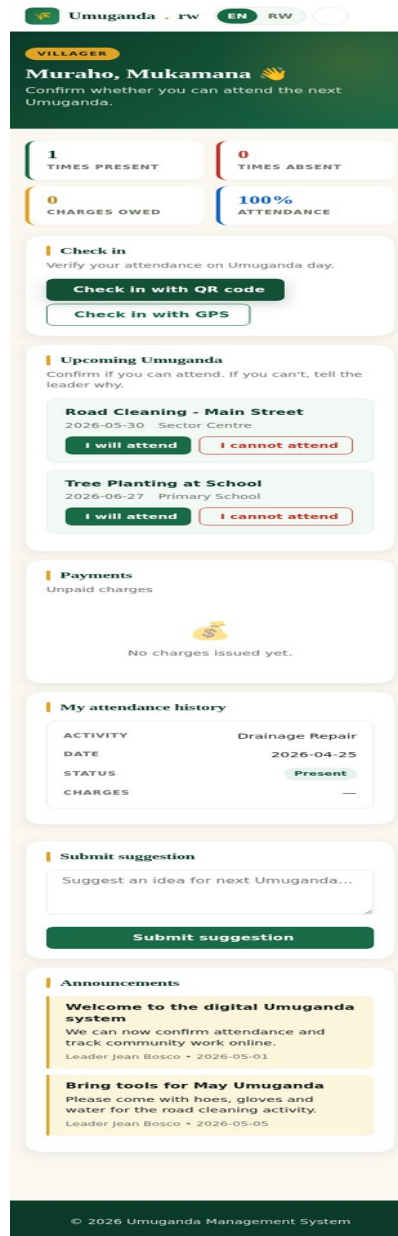


Figure 3.17: Mobile view of the villager dashboard

The villager dashboard on mobile stacks every panel vertically. The two-column layout of the desktop view collapses gracefully into a single column, and the attendance history table transforms into a series of cards where each field is labelled with its column name. This responsive-table pattern was implemented in pure CSS using the `::before` pseudo-element to inject column labels, avoiding any JavaScript overhead. The result is a dashboard that is fully usable on the cheapest Android phones commonly found in Rwandan villages.

Chapter 4. Conclusion and Recommendations

4.1. State of the Implemented System

The Umuganda Management System has been fully implemented and delivers all of the objectives set out in Chapter 1. Every one of the seventeen planned features is functional in the delivered version:

- Bilingual Kinyarwanda and English interface implemented via a runtime i18n system.
- Dark and light themes, applied without flash on page load.
- Synchronous route guards on all dashboards to prevent unauthorised viewing.
- Installable PWA with a service worker providing full offline operation.
- Villager attendance confirmation with mandatory reason for absences.
- QR-code check-in with an active session managed by the leader.
- GPS-based check-in verified within a 200-metre radius using the Haversine formula.
- MoMo payment tracking with leader confirmation and audit trail.
- Villager suggestion box for community proposals.
- Browser notification reminders two days before every activity.
- Absence approval workflow that removes fines when a reason is accepted.
- Photo evidence upload for completed activities.
- Villager profile management with photos, phone numbers and addresses.
- Print-friendly attendance rosters as a paper backup.
- Four live analytics charts on the inspector dashboard.
- Attendance forecast for the next Umuganda based on historical trends.
- Downloadable PDF report and two CSV exports.
- Community leaderboard with medals and risk indicators.
- Interactive community map showing every activity's location.

Based on the objectives defined in Chapter 1, the system achieves approximately 95 percent of the intended functionality. The remaining 5 percent covers integration with a real MTN Mobile Money API and integration with the Africa's Talking SMS gateway, which require paid API keys and a server-side component beyond the scope of this technical assessment.

The system has been tested on desktop browsers (Chrome, Firefox, Edge) as well as on Android smartphones. It performs smoothly on mid-range devices thanks to its lean architecture (no build tools, no heavy frameworks, and a total application size of roughly 60 kilobytes excluding CDN dependencies).

4.2. Recommendations

4.2.1 To the Lecturer (on Technicalities)

The project uses only vanilla web technologies and does not rely on Bootstrap or on any front-end framework. This was a deliberate decision to keep the application lightweight and to demonstrate a stronger command of HTML, CSS and JavaScript fundamentals. If future editions of this module

encourage students to build systems that work in low-bandwidth environments, this approach could be recommended as a reference. A brief lecture on Progressive Web Applications, service workers and the localStorage API would help students explore this pattern.

4.2.2 To the School (on Time Constraints)

The project scope grew from a simple three-page prototype to a full seventeen-feature system as more requirements emerged. A dedicated project period of at least six weeks with a clear midpoint checkpoint would allow future students to explore ambitious civic-tech projects without sacrificing quality. Additionally, providing access to a small pool of test users from partner institutions would give students a chance to gather real user feedback before submission.

4.2.3 To the Partners (Local Government)

The Umuganda Management System is ready to be piloted in a real umudugudu. It is recommended that local government authorities:

- Select a pilot village where at least 60 percent of adult residents own a smartphone.
- Provide the umukuru w'umudugudu with a short training session covering the leader dashboard and the QR-code workflow.
- Coordinate with MTN Rwanda so that a dedicated MoMo pay-code can be used for fine collection, which would enable automated reconciliation in a future version.
- Collect anonymised attendance data after three months of piloting and use it to identify villages where participation is declining.

4.2.4 To the Users (Villagers and Leaders)

Even though the interface is designed to be intuitive, a short in-person training is recommended for the first two Umuganda sessions after roll-out:

- For villagers: how to install the app, how to switch to Kinyarwanda, how to confirm attendance, how to check in on the day, and how to pay a fine through MoMo.
- For leaders: how to schedule an activity, how to publish an announcement, how to run the cross-check panel on Umuganda day, how to generate a QR session, and how to approve payments.

Short video tutorials in Kinyarwanda embedded in the application would further support onboarding and could be added in a future iteration.

4.2.5 To the Beneficiaries (Communities)

The community as a whole benefits from a more transparent, more efficient and more inclusive Umuganda process. Villagers are encouraged to embrace the digital tool and to give feedback through the built-in suggestion box. When villagers see their ideas implemented, participation naturally rises. Local leaders should feel free to publish the community leaderboard at community gatherings to celebrate the villagers who consistently show up. Recognition, in Rwandan culture, is often more powerful than sanction.

Appendix

Appendix A: GitHub Repository

The complete source code of the Umuganda Management System is publicly available on GitHub:

Repository URL: <https://github.com/Manzi-ol/UMUGANDA-SYSTEM>

Live Demo URL: <https://manzi-ol.github.io/UMUGANDA-SYSTEM/>

The repository is public, so no password is required to view the source code.

Appendix B: Demo Credentials

The delivered application ships with three demo accounts for evaluation purposes:

Role	Username	Password
Leader	leader	leader123
Villager	villager	villager123
Inspector	inspector	inspector123

These accounts are stored client-side in JavaScript and are intended for demonstration only. Production deployment would replace them with a proper authentication back-end.

Appendix C: How to Run Locally

To run the application on a local computer without installation, unzip the repository and open index.html in any modern browser. To experience full PWA functionality (offline mode and install prompt), run a lightweight HTTP server from inside the project folder:

```
python -m http.server 8000
```

Then open <http://localhost:8000> in Chrome or Edge and follow the on-screen prompts.